

Name _____ Date _____ Period _____

Parent Functions III: $1/x$ and Asymptotes

Algebra II Honors | Unit 1 | Day 6 | TEK A2.2.A

Objective: I can graph the reciprocal parent function, describe its end behavior precisely, prove it is odd, and locate vertical asymptotes of related rules from the denominator.

Vocabulary

Asymptotic behavior — how a graph behaves near a value it gets closer and closer to but never _____.

Vertical asymptote — a vertical line the graph _____, located where the function is _____.

Horizontal asymptote — describes _____. The graph gets closer and closer to it, without _____.

End behavior notation — "as $x \rightarrow \infty$, $f(x) \rightarrow 0$ " is read: as x grows _____, the output approaches 0.

Part 1 — Build the Table

x	-4	-2	-1	-1/2	0	1/2	1	2	4
$f(x) = 1/x$									

Why is $x = 0$ excluded, and what is the domain in set notation?

Division by zero is undefined, so _____

Part 2 — End Behavior, Precisely

Complete both patterns, then write the behavior in words.

x	1	0.1	0.01	0.001
$1/x$				

x	1	10	100	1000
$1/x$				

As $x \rightarrow 0$ from the right, $f(x) \rightarrow$ _____ As $x \rightarrow 0$ from the left, $f(x) \rightarrow$ _____

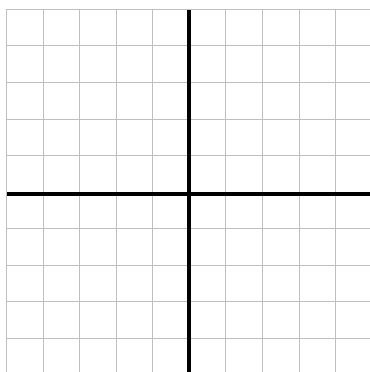
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____ As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

So the asymptotes are $x =$ _____ and $y =$ _____. Does the graph ever reach $y = 0$? _____

Part 3 — Graph and Prove

Each square is 1 unit. Two branches - do not connect them across the y -axis.

$f(x) = 1/x$, with both asymptotes dashed.



Prove f is odd.

$f(-x) =$ _____

so $f(-x) =$ _____

Quadrants used: _____

Intercepts: _____

Max or min: _____

Why does having no intercepts follow from the asymptotes?

The axes _____ the asymptotes, and a graph never reaches its asymptotes.

Part 4 — Asymptotes of Related Rules

Reason from the denominator. Where it is zero, there is a vertical asymptote.

Function	Denominator = 0 when	Vertical asymptote(s)	Domain
$1/(x - 3)$	_____	_____	_____
$1/(x^2 - 4)$	_____	_____	_____
$1/(x^2 + 1)$	_____	_____	_____

The last row has no vertical asymptote at all. Explain why:

$x^2 + 1$ is _____, so it is never zero for a real x .

Part 5 — Going Further

1. True or false, with a reason: a graph may cross a horizontal asymptote. _____

2. Does $1/x$ itself ever cross $y = 0$? _____ Is that a contradiction with item 1?

3. Solve $1/x = x$. Then say what the solutions mean on the graph.

$x =$ _____ — _____

Exit Ticket

Give the domain and both asymptotes of $f(x) = 1/(x + 5)$, and state its end behavior as $x \rightarrow \infty$.

Domain _____ VA _____ HA _____

End behavior: as $x \rightarrow \infty$, $f(x) \rightarrow$ _____